3577. Count the Number of Computer Unlocking Permutations

Difficulty: Medium

Link: https://leetcode.com/problems/count-the-number-of-computer-unlocking-permutations/description/?envType=daily-question&envId=2025-12-10

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Problem:

You are given an array `complexity` of length `n`.

There are `n` **locked** computers in a room with labels from 0 to `n - 1`, each with its own **unique** password. The password of the computer `i` has a complexity `complexity[i]`.

The password for the computer labeled 0 is **already** decrypted and serves as the root. All other computers must be unlocked using it or another previously unlocked computer, following this information:

- You can decrypt the password for the computer `i` using the password for computer `j`, where `j` is **any** integer less than `i` with a lower complexity. (i.e. `j < i` and `complexity[j] < complexity[i]`)
- To decrypt the password for computer `i`, you must have already unlocked a computer `j` such that `j < i` and `complexity[j] < complexity[i]`.

Find the number of permutations of `[0, 1, 2, ..., (n - 1)]` that represent a valid order in which the computers can be unlocked, starting from computer 0 as the only initially unlocked one.

Since the answer may be large, return it **modulo** $10^9 + 7$.

Note that the password for the computer **with label** 0 is decrypted, and *not* the computer with the first position in the permutation.

Example 1:

Input: complexity = [1,2,3]

Output: 2

Explanation:

The valid permutations are:

- [0, 1, 2]
  - Unlock computer 0 first with root password.
  - Unlock computer 1 with password of computer 0 since `complexity[0] < complexity[1]`.
  - Unlock computer 2 with password of computer 1 since `complexity[1] < complexity[2]`.
- [0, 2, 1]
  - Unlock computer 0 first with root password.
  - Unlock computer 2 with password of computer 0 since `complexity[0] < complexity[2]`.
  - Unlock computer 1 with password of computer 0 since `complexity[0] < complexity[1]`.

Example 2:

Input: complexity = [3,3,3,4,4,4]

Output: 0

Explanation:

There are no possible permutations which can unlock all computers.

Constraints:

- `2 <= complexity.length <= 10^5`
- `1 <= complexity[i] <= 10^9`